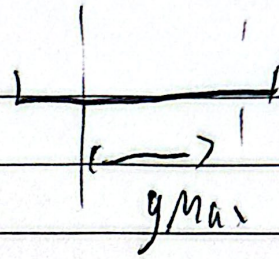
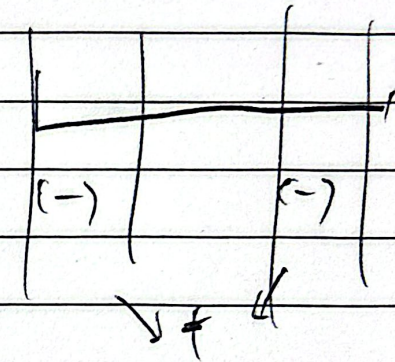


max subarray: 

Circular Subarray: 

Claim: If the max subarray doesn't bring the max, it'll come from total - gmin, maybe the biggest

Proof: For max'll always be in the form of prefix + suffix, the dual question is finding the min subarray. (2)

Claim: Max value ^{must be} ~~doesn't~~ in the middle when $g_{Max} \leq 0$

Proof: ~~In this case, $g_{min}/total \leq g_{Max} \leq 0$, $g_{min} \leq g_{Max} \leq 0$~~

1° In this case, $\forall num \leq 0$, otherwise, $\exists num > 0$, take $g_{Max} = num > 0$

2° By def, $g_{Min} \leq total$, but $\forall num \leq 0$, $total \leq g_{Min}$
 $\Rightarrow g_{Min} = total \leq g_{Max}$.